Technical Report: AuraFit — Evolutionary Multi-Objective Scheduling

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1. Abstract

Designing effective weekly training programs requires balancing multiple competing factors, including exercise selection, fatigue management, time constraints, and biological recovery [cite: 115]. This project formulates workout planning as a formal constrained scheduling problem solved through Evolutionary Computation. We propose AuraFit, a Genetic Algorithm (GA) engine that replaces myopic greedy heuristics with a global search approach. By evaluating entire 7-day microcycles simultaneously, the system eliminates "look-ahead blindness" and improves schedule quality. Experimental results demonstrate that AuraFit achieves a peak fitness score of 135.46, vastly outperforming baseline greedy methods (~38.40) while maintaining 100% adherence to strict physiological safety caps [cite: 119].

2. Introduction & Motivation

In traditional fitness environments, programming relies heavily on the intuition of coaches, which is often subject to heuristic bias [cite: 121]. However, exercise selection is highly interdependent; a neurologically demanding movement like a Barbell Squat on Monday creates a physiological lockout for similar movements for 48 to 72 hours [cite: 122]. This cascading decision tree closely mirrors the Resource-Constrained Project Scheduling Problem (RCPSP) [cite: 125]. AuraFit addresses this by applying a formal algorithmic framework to reduce bias and provide a systematic method for balancing weekly workloads and maximizing movement diversity [cite: 126].

3. System Architecture and Algorithmic Design

The optimization pipeline operates through an evolutionary search rather than a sequential constructive heuristic [cite: 129]. This allows the system to escape the local optima that frequently trap greedy algorithms [cite: 136].

- **Evolutionary Core (Genetic Algorithm):** The engine utilizes a nested map genotype representing a 7-day microcycle. It employs tournament selection, two-point crossover (swapping whole training days), and stochastic mutation to navigate the search space.
- **Portable Data Layer (SQLite):** To ensure cross-platform compatibility and ease of deployment for the engineering defense, the data layer was migrated from SQL Server to a portable SQLite database (exercise_vault.db).
- **Heuristic Judge (Evaluator):** The fitness function evaluates chromosomes across a weighted sum of metrics, including movement taxonomy, volume density, and fatigue variance.

4. Constraint Modeling and Domain Knowledge

The core of the scheduler relies on translating qualitative sports science principles into quantitative data structures [cite: 140].

- **Hard Constraints (Safety Caps):** Each training session is capped at 60 minutes and 80 fatigue units to manage acute workload, preventing overtraining [cite: 145].
- **The "Stick and Carrot" Logic:** To prevent the GA from avoiding work to minimize fatigue, a "Density Reward" is applied for sessions utilizing >=80% of the time cap, incentivizing high-quality, efficient stimulus.
- **Circular Recovery Windows:** The system enforces a 48-to-72 hour spacing between similar categories [cite: 147]. Critically, circular distance math is used to calculate recovery between Sunday (Day 6) and Monday (Day 0).
- **Biomechanical Taxonomy:** The engine seeks to satisfy minimum weekly frequencies across nine foundational movement patterns, including Squat, Hinge, Push, Pull, Lunge, Core, and Carry [cite: 148].

5. Experimental Results

AuraFit was tested against Random and Greedy baselines across 30 randomized trials to ensure statistical stability [cite: 151]. The GA demonstrated O(N) linear scaling relative to population growth.

Algorithm	Mean Total Score	Fatigue Balance (Variance)
Greedy Baseline	~38.40	Poor (Front-loaded)
Random Valid	~25.10	High Variance
AuraFit (GA)	135.46	Optimal (Normalized)

As illustrated, while the greedy algorithm executed rapidly, it exhibited poor fatigue distribution by front-loading the week [cite: 159]. AuraFit achieved a superior score by effectively balancing the assigned training load across all seven days.

6. Future Work & Conclusion

Future iterations will transition the data model from "Flat Fatigue" to Undulating Periodization (Fleck & Kraemer, 2014) [cite: 179]. This will involve oscillating daily fatigue caps to better represent multi-week macrocycle planning [cite: 180].

This project demonstrates that training design is a constrained optimization problem. The application of evolutionary computation is essential for escaping local optima and achieving a truly balanced physiological workload [cite: 183].

7. References

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